

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

1. Connector & input protection

Field connector, 2-stage bidirectional TVS surge protection, chassis isolation

File: connector.kicad_sch

2. Power

eFuse (reverse/OV/UV), 60V buck to 5V, LDO 3.3V

File: power.kicad_sch

3. MCU

STM32G0B1CCT3, SWD, reset, internal temperature sensor

File: mcu.kicad_sch

4. Measurement

Capacitive humidity sensor (PCAP04) and Pt1000 2-wire (ADS1220)

File: measurement.kicad_sch

5. Analog outputs

2x DAC8760 V/I output with TPS26611 miswiring protection

File: analog_out.kicad_sch

6. RS-485

THVD2450 +/-70V fault-protected transceiver

File: rs485.kicad_sch

DOTECH HMT500 – Oil Moisture Transmitter – Schematic v0.6

v0.5: J1 = JST GH 1.25 mm 8P side entry (SM08B-GHS-TB), harness W-2 to M12 8P; J5 chassis spring contact (was M12 shell pin).

v0.4: J3 = JST SH 1.0 mm 4P (SM04B-SRSS-TB side entry, v0.5) for sensor harness W-1 direct from HTX99R (no feedthrough, mech Rev F); Pt1000 2-wire (Kelvin split at J3 pads).

v0.3: MCU STM32G0B1CCT3 (LQFP48, -40..125 C, in-house part); TPS26600PWP official pinout; single axial PCB 57x23 (mech Rev C).

Signal flow: J1 field connector -> protection -> eFuse -> 5 V buck -> 3.3 V LDO; sensor head J3 -> PCAP04 / ADS1220 -> MCU -> DAC8760 x2 / THVD2450 -> J1

Outputs: RS-485 Modbus RTU + 2x analog (4-20 mA / 0-20 mA / 0-10 V / 0-5 V selectable). Supply 12-30 V DC.

Protection target: any pin pair +/-30 V continuous (miswiring), surge +/-1 kV, ESD +/-8 kV contact.

Power symbols: GND, +3V3, +3V3A (analog 3.3 V), +5V, VIN_P (protected input), CHASSIS. Inter-sheet signals: global labels.

v0.6: parts cleanup (-16: pull-ups R11-15, LED D3/R16, RS-485 R60/61 D60/61, FB2 C19 C11 C2 C6); PCB temperature = ADS1220 + MCU internal sensors.

VERIFY: LMR36006, PCAP04, DAC8760, TPS26611 pin numbers are placeholders (datasheets not accessible when drawn). TPS2660: RTN link R22 to confirm.

TBD footprints: J5 chassis contact, CMC, GDT, buck inductor, LMR36006, PCAP04, DAC8760, TPS26611.

표기 규칙: 부품번호 = 굵은 글자 (R1, U4) / 부품값 = 보통 글자 (10k, DAC8760) / 전원 네트 = 기울임 글자 (+3V3, VIN_P)

신호 네트 = 테두리 있는 라벨 (SPL_SCK, OUT1_EXT) - 네트 이름은 부품번호·부품명과 겹치지 않게 지음

글자 겹침: 생성기가 부품 몸체·핀·선·라벨·메모와 겹치지 않는 자리에 자동 배치하고, 겹치면 생성 실패로 처리

VERIFY=YES parts: pin numbers LMR36006/PCAP04/DAC8760/TPS26611; TPS2660 R22

Design notes: docs/hw/circuit-design.md

Oil moisture transmitter HMT500 – RS-485 + 2x V/I analog output

DOTECH Co., Ltd.

Sheet: /

File: HMT500.kicad_sch

Title: HMT500 Top

Size: A3Date: 2026-09-29Rev: 0.6

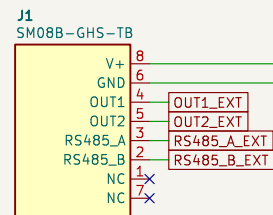
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4Id: 1/7

FIELD CONNECTOR

현장 커넥터

J1 JST GH 8P -> W-2 -> M12 8P

EE364 pinout, pin n = pin n



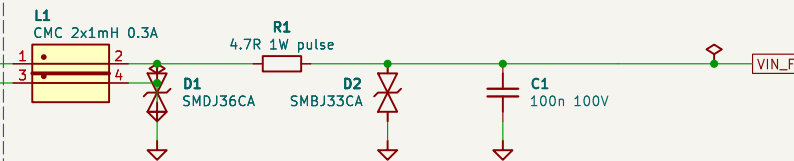
Pins 1, 7: not connected

GH mounting pads (MP): PCB only

INPUT SURGE / REVERSE-POLARITY PROTECTION

입력 서지·역극성 보호

TVS bidirectional: -30 V miswiring must not conduct



SMDJ 3 kW (1st) -> R1 -> SMBJ (2nd) -> eFuse on sheet Power

J5
Chassis contact

CHASSIS

CIRCUIT GND <--> CHASSIS (floating)

회로 GND-외함 분리



PCB-housing creepage >= 2 mm

GDT conducts only on line-to-ground surge

VERIFY=YES parts: pin numbers LMR36006/PCAP04/DAC8760/TPS26611; TPS2660 R22

Design notes: docs/hw/circuit-design.md

Oil moisture transmitter HMT500 - RS-485 + 2x V/I analog output

DOTEC Co., Ltd.

Sheet: /1. Connector & input protection/

File: connector.kicad_sch

Title: HMT500 Connector & input protection

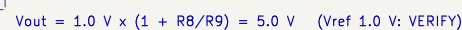
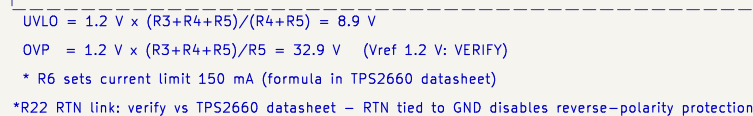
Size: A4

Date: 2026-09-29

Rev: 0.6

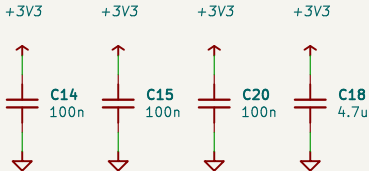
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4

Id: 2/7



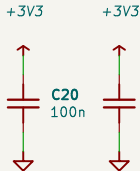
Size: A3	Date: 2026-09-29	Rev: 0.6
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 3/7

MCU DECOUPLING



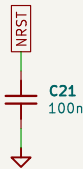
100 nF at VDD, VBAT, VREF+ (C20); 4.7 uF bulk

MCU 바이패스



RESET

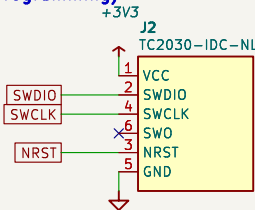
리셋



Fault lines: MCU pull-ups
(PB13-15, PC6-7 internal)

SWD (production programming)

생산 프로그래밍



MCU NOTES (STM32G0B1)

MCU 설정 메모

BOOT0 shares PA14/SWCLK - no pull-down.

Option bytes: nBOOT_SEL=1, nBOOT0=1

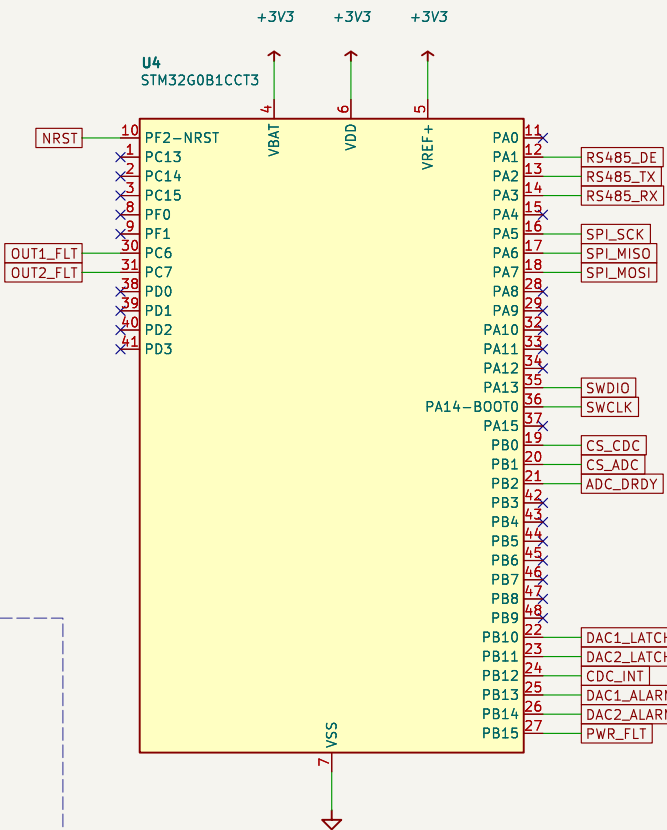
-> boot from flash (RS-485 bootloader 0x08000000).

VDDA = VDD (LQFP48); VREF+ tied to +3V3 (C20).

PCB temp (mid): internal sensor, ADC ch TS + TS_CAL.

Cortex-M0+: no SWO (J2 pin 6 NC).

Same die as DP2000 (G0B1CCT6); T3 = -40..125 C.



VERIFY=YES parts: pin numbers LMR36006/PCAP04/DAC8760/TPS26611; TPS2660 R22
Design notes: docs/hw/circuit-design.md
Oil moisture transmitter HMT500 - RS-485 + 2x V/I analog output

DOTECH Co., Ltd.

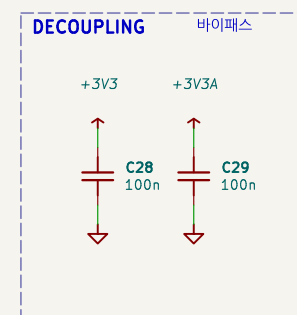
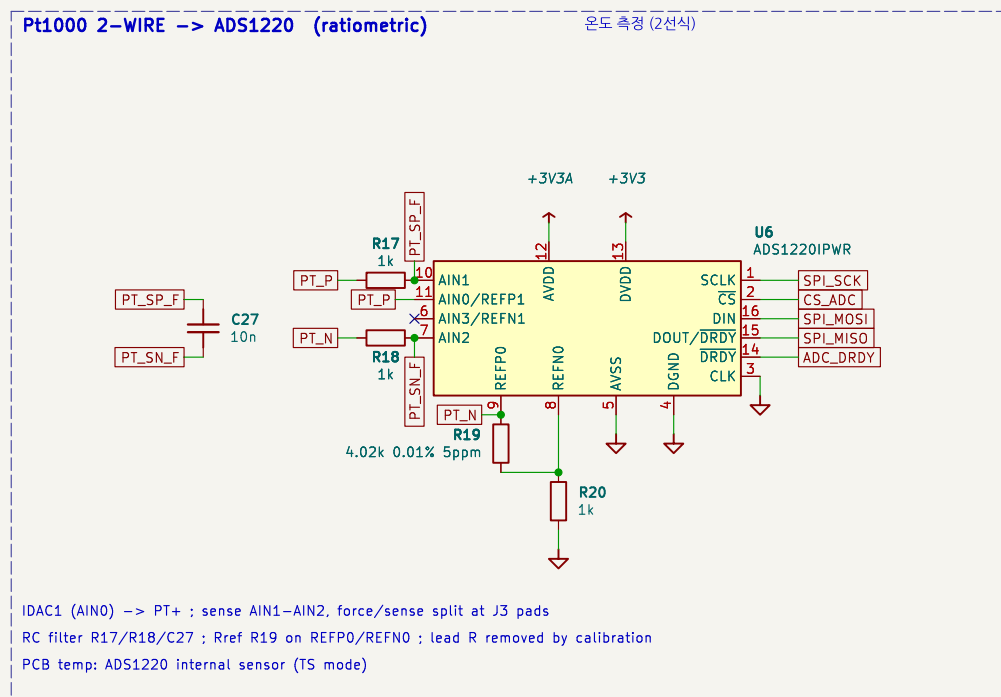
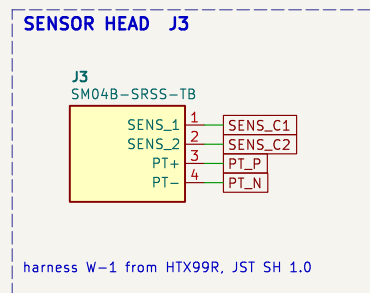
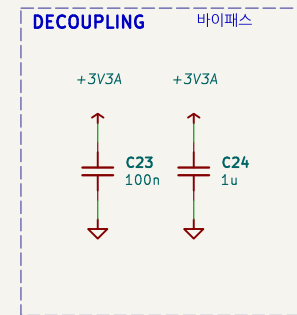
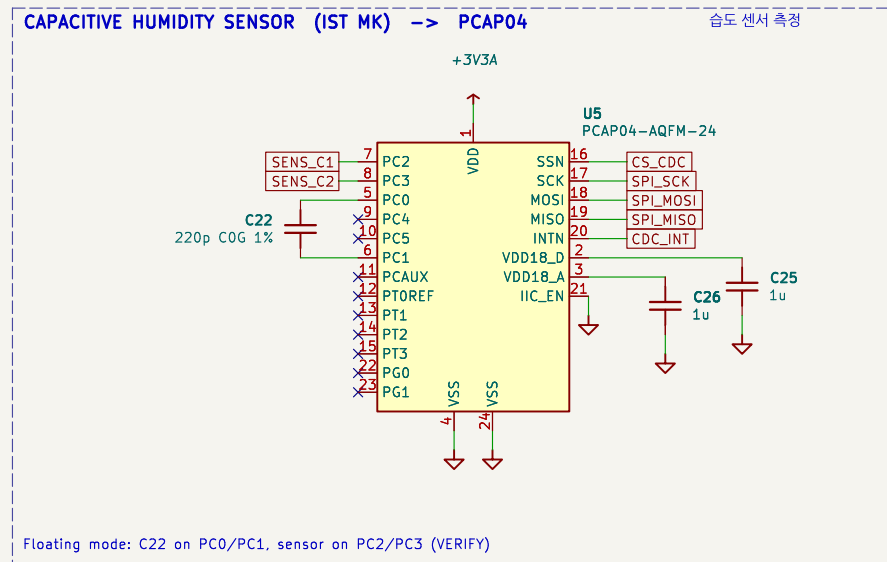
Sheet: /3. MCU/
File: mcu.kicad_sch

Title: HMT500 MCU

Size: A3 Date: 2026-09-29
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4

Rev: 0.6

Id: 4/7



VERIFY=YES parts: pin numbers LMR36006/PCAP04/DAC8760/TPS26611; TPS2660 R22
Design notes: docs/hw/circuit-design.md
Oil moisture transmitter HMT500 - RS-485 + 2x V/I analog output
DOTECH Co., Ltd.

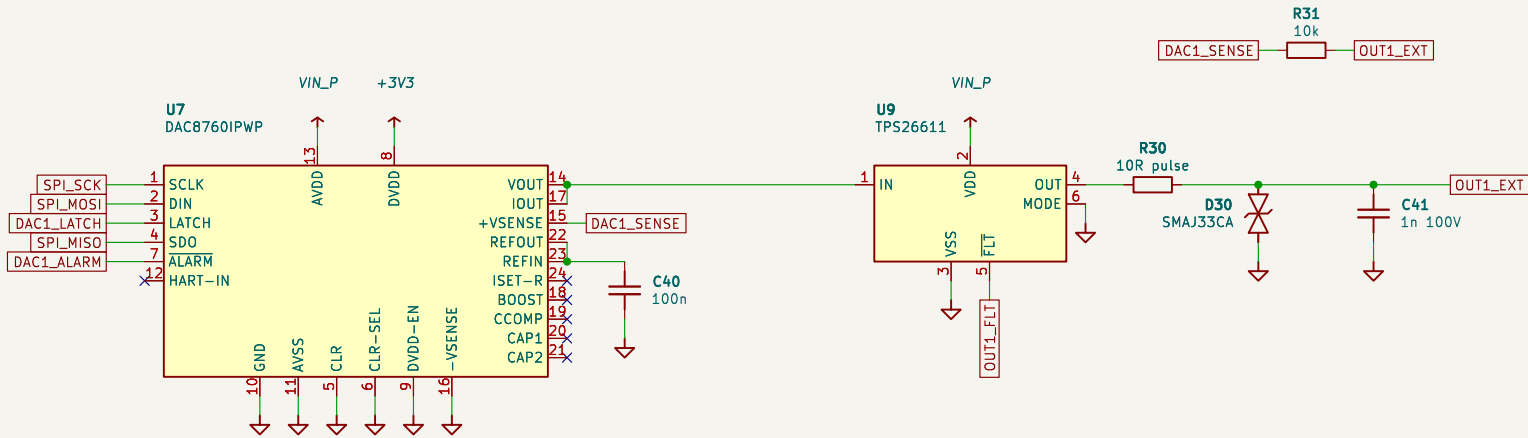
Sheet: /4. Measurement/
File: measurement.kicad_sch

Title: HMT500 Measurement

Size: A3	Date: 2026-09-29	Rev: 0.6
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 5/7

ANALOG OUTPUT CH1 - V/I selectable, miswiring protected -> connector pin 4

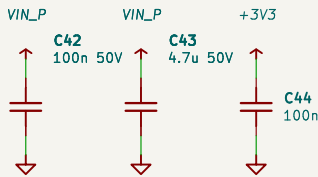
아날로그 출력 1



VOUT and IOUT combined (per TI TIDUBK2 - VERIFY). +VSENSE after R30 via R31.

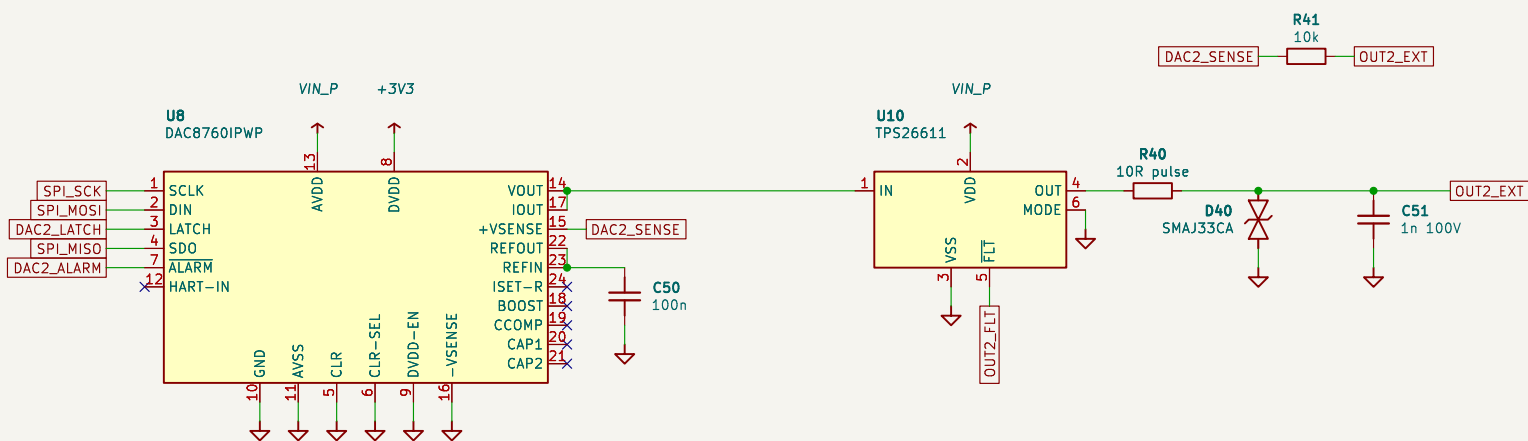
CH1 DECOUPLING

채널 1 바이패스



ANALOG OUTPUT CH2 - V/I selectable, miswiring protected -> connector pin 5

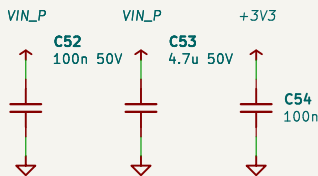
아날로그 출력 2



VOUT and IOUT combined (per TI TIDUBK2 - VERIFY). +VSENSE after R40 via R41.

CH2 DECOUPLING

채널 2 바이패스



DAC SELECTION (decision 2026-09-26: fit DAC8760 now; final part after output spec is fixed)

DAC 선정 (결정 #10)

U7/U8 fitted: DAC8760 (16-bit, current + voltage output). Same package/pinout family - drop-in alternates, no layout change:

DAC7760 12-bit, current + voltage (cost-down if V/I selectable output is kept)

DAC8750 16-bit, current only 4-20/0-20/0-24 mA (EE364-equivalent, 3-wire)

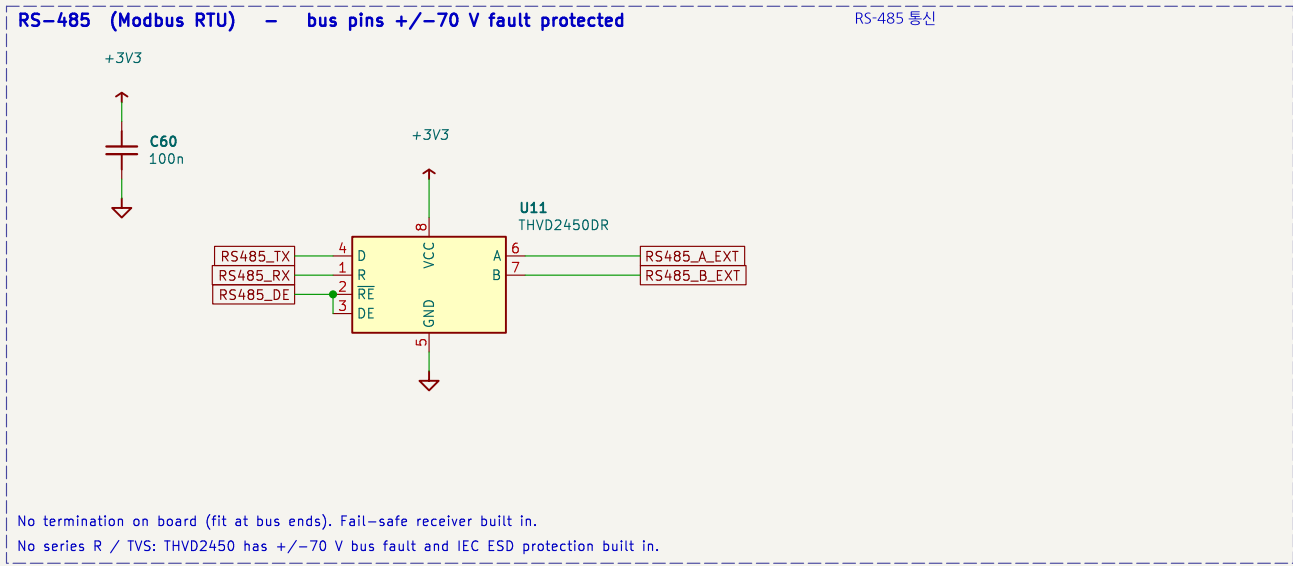
DAC7750 12-bit, current only (lowest cost; -5 uA/step vs EE364 2 uA)

Current-only (x750): VOUT/+VSENSE/-VSENSE functions absent - check pin handling in datasheet before fitting (VERIFY).

VERIFY=YES parts: pin numbers LMR36006/PCAP04/DAC8760/TPS26611; TPS2660 R22
Design notes: docs/hw/circuit-design.md
Oil moisture transmitter HMT500 - RS-485 + 2x V/I analog output
NOTECH Co., Ltd.
Sheet: /5. Analog outputs/
File: analog_out.kicad_sch

Title: HMT500 Analog outputs

Size: A3 Date: 2026-09-29 Rev: 0.6
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 Id: 6/7



VERIFY=YES parts: pin numbers LMR36006/PCAP04/DAC8760/TPS26611; TPS2660 R22
Design notes: docs/hw/circuit-design.md
Oil moisture transmitter HMT500 - RS-485 + 2x V/I analog output

DOTTECH Co., Ltd.

Sheet: /6. RS-485/
File: rs485.kicad_sch

Title: HMT500 RS-485

Size: A4	Date: 2026-09-29	Rev: 0.6
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4	Id: 7/7	